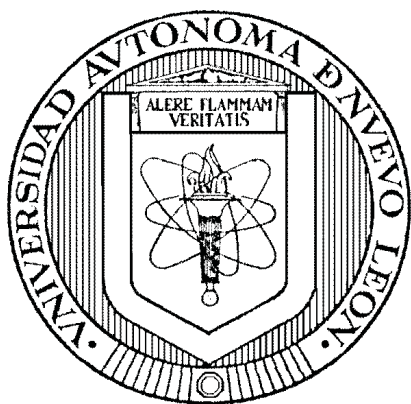


UNIVERSIDAD AUTÓNOMA DE NUEVO LEÓN
FACULTAD DE INGENIERÍA MECÁNICA Y ELÉCTRICA
SUBDIRECCIÓN DE ESTUDIOS DE POSGRADO



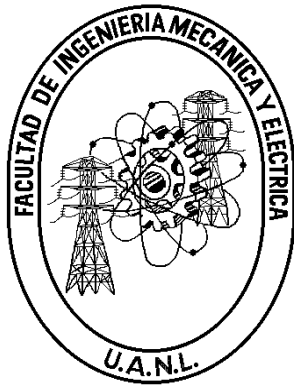
YIELD-AWARE HIGHER-ORDER TRAJECTORY OPTIMIZATION FOR
STEP-AND-SCAN WAFER STAGES

POR
ROBERTO TREVIÑO CERVANTES

COMO REQUISITO PARA OBTENER EL GRADO DE
MAESTRÍA EN CIENCIAS DE LA INGENIERÍA ELÉCTRICA

OCTUBRE 2026

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Los miembros del Comité de Tesis recomendamos que la tesis “**Yield-Aware Higher-Order Trajectory Optimization for Step-and-Scan Wafer Stages**” realizada por el(la) alumno(a) **Roberto Treviño Cervantes**, con número de matrícula 1915003, sea aceptada para su defensa como opción al grado de **Maestría en Ciencias de la Ingeniería Eléctrica**.

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A ... por su apoyo ... ETC

Agradecimientos

Agradezco a ... blah blah blah.

Abstract

El termino *caotico* es asignado a la clase de dinamicas en sistemas fisicos y matematicos deterministicos cuyo historial en el tiempo tiene una sensible dependencia a las condiciones iniciales. Este tipo de dinamica se presenta frecuentemente en la naturaleza en los sistemas dinamicos no lineales. La sensibilidad a las condiciones iniciales que presentan estas dinamicas es causa de impredecibilidad, tal como en los modelos climaticos, por lo cual es deseable en muchas ocasiones eliminar las dinamicas caoticas con la finalidad de trabajar con sistemas cuyas trayectorias no sean tan sensibles a pequenas variaciones en las condiciones iniciales o en los parametros...

Symbols and notation

$\square$	Fin de una prueba.
$\dot{x}, \dot{y}$	El punto es utilizado para denotar la derivada con respecto del tiempo, ejemplo $\dot{x} = x^{(1)} = dx/dt$
$\hat{x}, \hat{z}$	El circunflejo es utilizado para denotar la estimacion de una variable mediante observadores.
A^T, x^T	La transpuesta de la matriz A y el vector x .
$\mathbb{R}$	El campo de los numeros reales.
$\mathbb{C}$	El campo de los numeros complejos.
∞	Infinito.
$rango(A)$	El rango de la matriz A .
$\triangleq$	Igual por definicion.
$\approx$	Aproximadamente igual.
$\equiv$	Identico.
$dist\{x, y\}$	La distancia entre los puntos x y y , en algun espacio definido.
$\inf x$	El <i>infimum</i> de x .
$\sup x$	El <i>supremum</i> de x .
$\ x\ $	Una norma de x .
$\lim_{a \rightarrow b} x$	El limite de x cuando a tiende a b .
$\ln$	Logaritmo natural (en base e) de x .
$\log x$	Logaritmo en base 10 de x .
$\dim X$	La dimension del conjunto X .
$\mathcal{C}^\infty$	El conjunto de la funciones continuamente diferenciables.
A^{-1}	Inversa de A .
j	Unidad de los numeros imaginarios, $j = \sqrt{-1}$.
$\partial F / \partial x$	Derivada parcial de la funcion F con respecto a la variable x .
$L_f h$	Derivada Lie de h a lo largo del campo vectorial f .
<i>Cursivas</i>	El uso de cursivas sirve para denotar terminos en un idioma diferente al Espanol.
Negritas	El uso de negritas sirve para denotar terminos tecnicos.

Table of Contents

Agradecimientos	iv
Abstract	v
Symbols and notation	vi
Table of Contents	vii
List of Tables	viii
List of Figures	ix
1 Introduction	1
2 Model	6
2.1 Introduction	6
2.2 Dynamics	9
2.2.1 For the base frame	13
2.2.2 For each chain	14
2.2.3 State-space formulation	15
3 Control Objectives	20
3.1 Introduction	20
4 Methodology	21
4.1 Introduction	21
4.2 Simulation	21
Conclusion	27
A Full-State Space Model	29
A.1 State-Space Equations for the Wafer Chain	29

List of Tables

2.2.1 State-space mapping for the global concatenated system vector	17
4.2.1 Mapping of MuJoCo XML bodies to the model components	22
4.2.2 MuJoCo XML tag functions in the scanner simulation	23
4.2.3 Wafer chain stiffness (K) and damping (C) parameters	23
4.2.4 Reticle chain stiffness (K) and damping (C) parameters	24
4.2.5 Optics chain stiffness (K) and damping (C) parameters	24

List of Figures

2.1	Lumped mass-spring-damper system of the three chains vertically stacked on the common base frame	8
4.1	Screenshot of the MuJoCo simulation in the midst of the scanning motion, showing the wafer and the optics chain under the metrology frame	25

Chapter 1

Introduction

The history of modern electronics is inseparable from the history of lithography. Since the independent invention of the integrated circuit (IC) by Jack Kilby at Texas Instruments and Robert Noyce at Fairchild Semiconductor in 1958, Moore’s Law has governed the relentless progression of the semiconductor industry: the density of components on a chip doubles approximately every two years⁶. This exponential trend has translated into dramatic reductions in cost alongside equally dramatic increases in performance. The price of one gigabyte of NAND flash memory, for instance, fell from over \$1000 at the start of this millennium to below \$0.30 by the mid-2010s³, while the minimum transistor feature size has shrunk from hundreds of nanometres to just a few tens of nanometres. The engine of this revolution is *photolithography* — the process by which the intricate patterns of a transistor, a wire, or a memory cell are projected onto a silicon wafer with nanometre-scale fidelity.

The fabrication of a modern IC consists of up to 30 sequential process cycles, each defining one distinct layer of the device⁶. A monocrystalline silicon wafer — a thin disc of up to 300 mm in diameter — serves as the substrate throughout this process. Each cycle begins with a chemical pre-treatment, such as the thermal growth of a thin oxide film for electrical isolation. A photosensitive material called *resist* is then deposited onto the wafer surface. An optical exposure system projects the pattern for the current layer — defined in a chromium film on a transparent

quartz plate known as the *reticle*, or *photo-mask* — onto the resist. After development, the regions that received light are selectively dissolved (in the case of a positive resist), exposing the underlying material. This material then undergoes one of several process steps. *Plasma etching* selectively removes oxide or semiconductor material to define trenches and isolation regions. *Ion implantation* locally modifies the electrical properties of the silicon to form transistor junctions and doped wells. *Physical or chemical vapour deposition* lays down thin films of conductive metal — typically tungsten or copper — to form the interconnecting wiring that links the millions of functional elements stacked across all layers. After processing, the remaining resist is stripped and the cycle repeats for the next layer, until all layers are complete and the wafer is diced and packaged.

In the earliest days of the industry, the reticle was brought into direct physical contact with the resist-coated wafer during exposure — a technique known as *contact lithography*. While simple, this approach had two fundamental shortcomings. First, repeated physical contact progressively contaminated and damaged the delicate reticle pattern, necessitating its costly and time-consuming replacement. Second, a 1:1 projection offers no optical resolution advantage: any defect present on the reticle is transferred at full size onto the wafer. These limitations motivated a transition to *projection optical lithography*, in which a reduction lens images the reticle pattern onto the wafer from a safe working distance. The projection lens introduces a demagnification factor of four, meaning that features on the reticle are printed four times smaller on the wafer, immediately relaxing the demands on reticle fabrication. The minimum printable feature size, known as the *critical dimension* (CD), is governed by⁶:

$$\text{CD} = k \frac{\lambda}{\text{NA}}, \tag{1.0.1}$$

where λ is the illumination wavelength, NA is the numerical aperture of the projection lens, and k is a process-dependent factor. Reducing CD requires a shorter wavelength, a higher NA, and process improvements that lower k . This relation has driven a continuous push toward shorter wavelengths — from mercury arc-lamp sources at 436 nm and 365 nm, through excimer

lasers at 248 nm and 193 nm, to the extreme ultraviolet (EUV) sources at 13.5 nm now entering production.

Achieving a high numerical aperture, however, comes with a practical constraint. Since $\text{NA} = n \sin \vartheta$, where ϑ is the half-angle of the marginal ray accepted by the lens, attaining values of NA close to one in air requires the lens to subtend a very wide solid angle. Constructing a lens element that simultaneously covers the full area of a 300 mm wafer at high NA would require an optical element of impractical size and cost⁶. The industry's solution was to accept a smaller *exposure field* — typically 26×32 mm, corresponding to a single IC die — and expose the wafer in successive steps: after each die is exposed, the wafer stage repositions the wafer to the next die location and the exposure is repeated. This architecture defines the *wafer stepper*. In a stepper, each die is illuminated in a single, instantaneous full-field flash, with the stage brought to a complete stop before exposure commences.

While the stepper represented a decisive step forward, exposing an entire die simultaneously subjects the full field to any non-uniformities in the illumination intensity or lens aberrations. The *scanner* addresses this by replacing the full-field flash with a *scanning exposure*: the illumination system reveals only a narrow rectangular slit of the reticle, and the reticle stage sweeps the mask through this slit while the wafer stage moves synchronously in the opposite direction, maintaining the 4:1 demagnification relationship throughout^{3;6}. Because the reticle moves at four times the speed of the wafer, it traverses the full reticle pattern in the same time that the wafer moves one die height. This scanning motion offers significant advantages over full-field exposure: it averages out illumination dose non-uniformities across the exposure field, improves imaging uniformity at the field edges, and allows the use of a physically larger reticle that is easier to write with electron-beam pattern generators. After each die is scanned, the wafer is stepped to the next die position and the scan repeats, defining the *step-and-scan* cycle that gives modern lithographic tools their name. A throughput of over 175 wafers per hour — corresponding to less than 0.2 s of exposure time per die — is not uncommon in state-of-the-art

tools³.

A critical consequence of the multi-layer nature of IC fabrication is the requirement that each new layer be positioned with nanometre-level accuracy relative to all preceding layers — a specification known as *overlay*. The functional layers of a chip are interconnected by millions of tiny conductive pillars called *vias*; a misregistration between consecutive layers of even a few nanometres can misalign a via, break an intended connection, or create an unintended short circuit, rendering the IC defective⁶. Overlay requirements scale roughly in proportion with the CD and currently sit below 4 nm for leading-edge nodes. Overlay is therefore one of the primary technology drivers in scanner design, and it is governed directly by the positioning accuracy of the wafer and reticle stages during the scanning exposure.

The commercial performance of a scanner is ultimately measured in wafers per hour (WPH), since a machine whose capital cost can exceed \$150 million must generate revenue continuously. Although the exposure time per die is fixed by the optics and the process, the time spent stepping and *settling* — the interval between the end of a step motion and the moment the stage has damped its residual vibrations to within the nanometre-level positioning specification — represents a significant and controllable fraction of the total cycle time³, which places the allowable settle time below 10 ms. Every millisecond saved in settling directly translates into more dies exposed per hour and thus more chips delivered per unit time, with a lower cost per chip. Reducing settling time is therefore a central objective in scanner control engineering.

It is worth noting that, while EUV lithography at 13.5 nm has entered high-volume production for leading-edge nodes below 10 nm, deep ultraviolet (DUV) optical lithography at 193 nm remains the dominant technology for *mature nodes* — technology generations at 28 nm and above. Mature-node chips are the backbone of automotive electronics, industrial controllers, power management devices, analogue circuits, and the vast majority of consumer electronics. The aggregate volume of wafers produced at mature nodes substantially exceeds that at leading-edge nodes, and is projected to remain so for the foreseeable future. Throughput improvements

in DUV step-and-scan tools therefore continue to have a large global impact.

The key lever for reducing settling time — and thus for increasing scanner throughput — is the generation of *reference trajectories*: motion profiles that command the stage from its current position to the next scanning start-point with minimal excitation of mechanical resonances and the shortest possible residual settling tail. A well-designed trajectory limits the energy injected into the structural eigenmodes of the machine, reduces induced vibrations in the sensitive metrology and optical systems, and enables the stage to reach nanometre-level positioning accuracy well within the allocated window. The design of such trajectories is a problem of direct industrial relevance, and forms the central subject of this work.

Chapter 2

Model

2.1 Introduction

As we established before, the accurate relative positioning between the reticle and the wafer is crucial for *step-and-scan* machines to produce functional ICs. To analyze these positioning requirements in a multi-body framework, Al-Rawashdeh et al² propose modeling the lithography scanner as a combination of three open-loop serial chains: the reticle chain ($j = 1$, blue), the optics chain ($j = 2$, purple), and the wafer chain ($j = 3$, red), the latter being of primary interest for the *step-and-scan* motion problem studied here.

The base frame serves as the common structural root for all three chains. In the reticle chain, a dual-stage architecture is employed to satisfy the competing requirements for travel range and precision. The long-stroke motor (LS) of the planar moving-coil type⁷ consisting of four coil actuators arranged in a platform shape which thus facilitates large-range travel (of at least 300mm) with moderate micrometer accuracy in 6-DOF. However, being directly mounted to the base frame, it is inherently susceptible to vibrations propagated from the rest of the machine.

To achieve nanometric tracking accuracy, a short-stroke motor (SS) is appended to the long-stroke carrier. Since travel in this stage is limited up to 1mm, it is realized with Lorentz actuators which provide contact-free, high-bandwidth positioning. Given that the Lorentz force depends entirely on the current within the actuator and is independent of the stator’s motion, it effectively decouples the end-effector from the base frame disturbances through magnetic suspension³. To minimize linearity errors and optimize force transfer the long-stroke stage carrying the stator must maintain the optimal magnetic gap to the magnets of the rest of the short-stroke actuator. This arrangement is replicated for the wafer chain.

The optics chain is attached to the base frame through a metrology frame which is in turn supported by pneumatic vibration isolators³. While the optics box includes adaptive optics with active elements, it is reduced in this framework to a single lens element. This allows the model to focus on machine compliance and minimizing structural vibration transfer rather than involving active optical control².

The physical connections between these stages, which constitute the primary transmission path for vibration, are represented by lumped-parameter spring-damper systems, according to the Kelvin-Voigt model. As its proponents state, this low-fidelity modeling choice is sufficient to study the dynamics between the multiple bodies during *step-and-scan* motion².

Figure 2.1 illustrates the structural topology and hierarchical arrangement within each chain. Horizontal Kelvin–Voigt elements $(K_{i,j}, C_{i,j})$ couple each stage laterally to the fixed structural reference depicted as a solid line in the color of the corresponding chain; in particular, elements (K_1, C_1) provide passive vibration isolation between $\mathcal{F}_0 \rightarrow \mathcal{F}_1$ through the solid green line that extends the base frame ($i = 1$) into the coupling to the first body ($i = 2$) in any j chain. The in-plane deviations $f_{i,j}$ represented by the horizontal blue double-headed arrows measure the relative displacement between each body. The dashed vertical centerline represents the nominal reference for $f_{i,j}^0$, when all relative coordinates are zero.

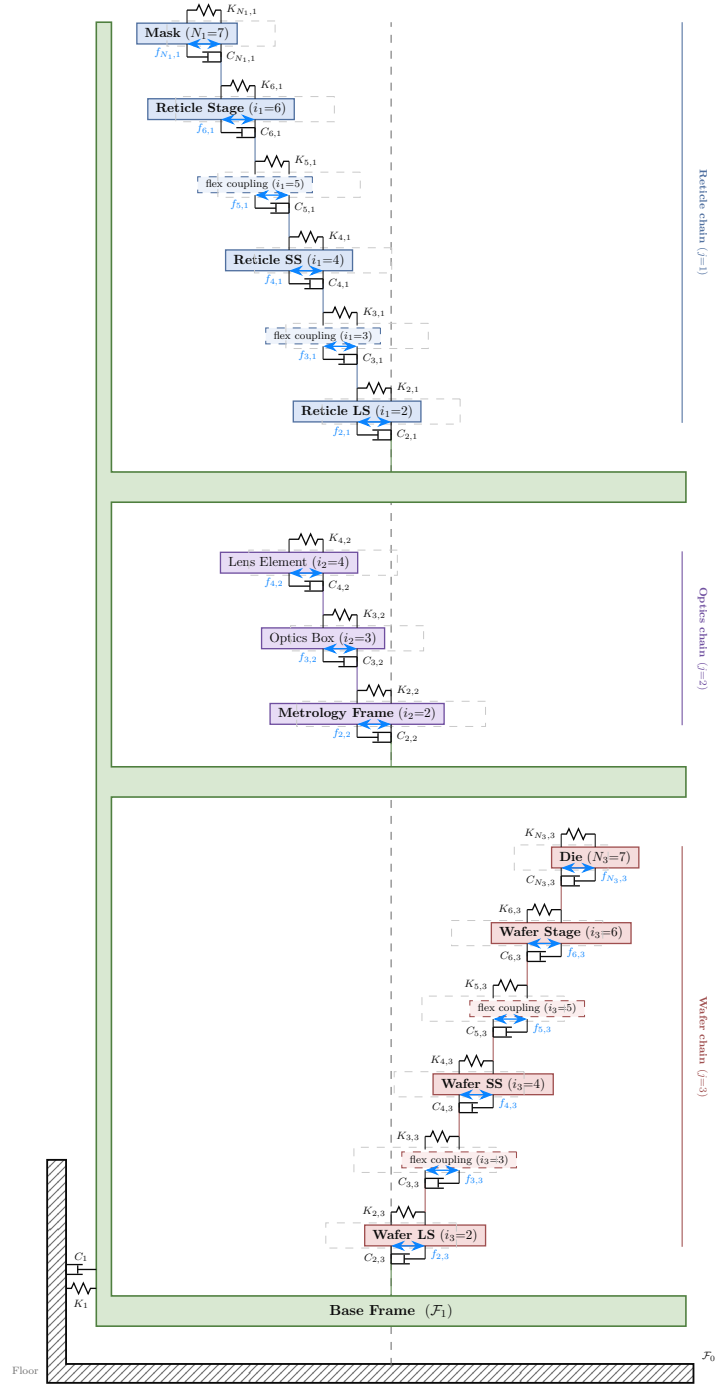


Figure 2.1: Lumped mass-spring-damper system of the three chains vertically stacked on the common base frame

2.2 Dynamics

The full 6-DOF model is here restricted to in-plane motions^{1,4}, since rotations about the x and y axis, as well as translation on z are used mainly for focusing. The generalized coordinate used to describe the configuration of the i th stage within the j th chain is

$$q_{i_j} = \begin{bmatrix} f_{i_j} \\ \theta_{z_{i_j}} \end{bmatrix} \in \mathbb{R}^{3 \times 1} \quad (2.2.1)$$

where $f_{i_j} = [f_{x,i_j}, f_{y,i_j}]^T \in \mathbb{R}^2$ collects the in-plane translational displacements of stage i relative to its parent body in chain j , and $\theta_{z_{i_j}} \in \mathbb{R}$ is the corresponding rotation about the out-of-plane z -axis. The joints between bodies within a chain are modeled using a spring-damper system².

The associated generalized mass-inertia matrix for each body is

$$M_{i_j} = \begin{bmatrix} m_{i_j} \mathbf{I}_2 & 0 \\ 0 & I_{z_{i_j}} \end{bmatrix} \in \mathbb{R}^{3 \times 3} \quad (2.2.2)$$

and the generalized stiffness and damping matrices $K_{i_j}, C_{i_j} \in \mathbb{R}^{3 \times 3}$ are block-diagonal, with a 2×2 translational block and a scalar rotational entry.

The absolute position $f_{i_j}^0$ of body i is expressed as the recursive sum of its relative displacement from its parent body along the chain from the floor frame $\mathcal{F}_0$. Thus, for bodies with $i \geq 2$ on any chain $j \in \{1, 2, 3\}$:

$$f_{i_j}^0 = f_{(i-1)_j}^0 + f_{i_j} \quad (2.2.3)$$

Since the rotation about the z -axis is a scalar, the angular position recurrence is simply additive:

$$\theta_{z_{i_j}}^0 = \theta_{z_{(i-1)_j}}^0 + \theta_{z_{i_j}} \quad (2.2.4)$$

From this, the translational velocity expression appears as:

$$\dot{f}_{i_j}^0 = \dot{f}_{(i-1)_j}^0 + \dot{f}_{i_j} + \omega_{(i-1)_j} \times f_{i_j} \quad (2.2.5)$$

Where $\omega_{(i-1)_j}$ is the angular velocity⁵ of $\mathcal{F}_{(i-1)_j}$ with respect to the floor, $\mathcal{F}_0$.

Taking the second derivative gives:

$$\begin{aligned}
\ddot{f}_{i_j}^0 &= \ddot{f}_{(i-1)_j}^0 + \ddot{f}_{i_j} + \omega_{(i-1)_j} \times \dot{f}_{i_j} + \omega_{(i-1)_j} \times f_{i_j} + \omega_{(i-1)_j} \times [\dot{f}_{i_j} + \omega_{(i-1)_j} \times f_{i_j}] \\
&= \ddot{f}_{(i-1)_j}^0 + \ddot{f}_{i_j} + \omega_{(i-1)_j} \times \dot{f}_{i_j} + \omega_{(i-1)_j} \times f_{i_j} + \omega_{(i-1)_j} \times \dot{f}_{i_j} + \omega_{(i-1)_j} \times [\omega_{(i-1)_j} \times f_{i_j}] \quad (2.2.6) \\
&= \ddot{f}_{(i-1)_j}^0 + \ddot{f}_{i_j} + 2 \left[\omega_{(i-1)_j} \times \dot{f}_{i_j} \right] + \omega_{(i-1)_j} \times f_{i_j} + \omega_{(i-1)_j} \times [\omega_{(i-1)_j} \times f_{i_j}]
\end{aligned}$$

Since we are concerned with planar movements in this work, we restrict the angular velocity of each body purely about the z -axis as $\omega = \dot{\theta}_z \hat{z}$, so that:

$$\omega_{(i-1)_j} = \dot{\theta}_{z(i-1)_j}^0 \hat{z}$$

We substitute the cross-products as matrix-vector products using the skew-symmetric matrix. In the 6-DOF model², it appears as:

$$[\omega] = \begin{bmatrix} 0 & -\omega_z & \omega_y \\ \omega_z & 0 & -\omega_x \\ -\omega_y & \omega_x & 0 \end{bmatrix}$$

Clearly, here we can set $\omega_x = 0$ and $\omega_y = 0$, thus $\omega = [0, 0, \dot{\theta}_z]^T$ and the matrix becomes:

$$[\omega] = \begin{bmatrix} 0 & -\omega_z & 0 \\ \omega_z & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Considering our 2D displacement vector, we only take the upper-left 2×2 sub-matrix, and factoring out the scalar $\dot{\theta}_{z(i-1)_j}^0$:

$$[\omega_{(i-1)_j}] = \dot{\theta}_{z(i-1)_j}^0 \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \quad (2.2.7)$$

such that $\omega_{(i-1)_j} \times f_{i_j} = [\omega_{(i-1)_j}] f_{i_j}$. Applying this matrix twice gives the centripetal term:

$$[\omega_{(i-1)_j}]^2 = - \left(\dot{\theta}_{z(i-1)_j}^0 \right)^2 \mathbf{I}_2 \quad (2.2.8)$$

so that $\omega_{(i-1)_j} \times [\omega_{(i-1)_j} \times f_{i_j}] = [\omega_{(i-1)_j}]^2 f_{i_j}$. Substituting and expanding (2.2.3) recursively

from the floor frame, the translational kinematics become:

$$f_{i_j}^0 = f_1^0 + \sum_{k=2}^i f_{k_j} \quad (2.2.9)$$

$$\dot{f}_{i_j}^0 = \dot{f}_1^0 + \sum_{k=2}^i \dot{f}_{k_j} + \sum_{k=2}^i [\omega_{(k-1)_j}] f_{k_j} \quad (2.2.10)$$

$$\ddot{f}_{i_j}^0 = \ddot{f}_1^0 + \sum_{k=2}^i \ddot{f}_{k_j} + \sum_{k=2}^i \left[2 [\omega_{(k-1)_j}] \dot{f}_{k_j} + [\dot{\omega}_{(k-1)_j}] f_{k_j} + [\omega_{(k-1)_j}]^2 f_{k_j} \right] \quad (2.2.11)$$

Since the rotation $\theta_{z_{i_j}}$ is a scalar, no Coriolis or centripetal cross-terms arise in the angular kinematics. Expanding (2.2.4) recursively from the floor frame gives the rotational analogues:

$$\theta_{z_{i_j}}^0 = \theta_{z_1}^0 + \sum_{k=2}^i \theta_{z_{k_j}} \quad (2.2.12)$$

$$\dot{\theta}_{z_{i_j}}^0 = \dot{\theta}_{z_1}^0 + \sum_{k=2}^i \dot{\theta}_{z_{k_j}} \quad (2.2.13)$$

$$\ddot{\theta}_{z_{i_j}}^0 = \ddot{\theta}_{z_1}^0 + \sum_{k=2}^i \ddot{\theta}_{z_{k_j}} \quad (2.2.14)$$

Thus the equation of the angular motion of the i th body in the j th chain w.r.t $\mathcal{F}_0$ is:

$${}^0I_j \dot{\omega}_j + ([\omega_j]) {}^0I_j \omega_j = m_j^0 \quad (2.2.15)$$

Under the planar restriction $\omega_x = \omega_y = 0$ the gyroscopic term vanishes like:

$$([\omega_{i_j}] {}^0I_{i_j}) \omega_{i_j} = \begin{bmatrix} 0 & -\dot{\theta}_z & 0 \\ \dot{\theta}_z & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} I_{xx} & 0 & 0 \\ 0 & I_{yy} & 0 \\ 0 & 0 & I_{zz} \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_z \end{bmatrix} = 0$$

Leaving only:

$$I_{z_{i_j}} \ddot{\theta}_{z_{i_j}}^0 = m_{z_{i_j}}^0 \quad (2.2.16)$$

where $I_{z_{i_j}}$ is the aggregated moment of inertia of all bodies from i to the end-effector N_j , referred to the z -axis of body i via Steiner's theorem, and $m_{z_{i_j}}^0 \in \mathbb{R}$ is the net applied moment acting on body i in chain j about the z -axis.

Steiner’s theorem and inertia aggregation. The parallel axis (Steiner’s) theorem states that the moment of inertia of a rigid body about any axis parallel to one through its center of mass (CoG) satisfies

$$I_z = \tilde{I}_z + m \|f\|^2, \quad (2.2.17)$$

where $\tilde{I}_z$ is the body’s own moment of inertia about the z -axis through its CoG, m is its mass, and $\|f\|^2 = f_x^2 + f_y^2$ is the squared in-plane distance between the two parallel axes. In a serial chain, the inertia of every downstream body must be shifted to the frame of the current body using (2.2.17). Following the characterization article² (Algorithm 1 therein), the aggregated inertia for the j th chain is computed recursively from the end-effector inward. In the planar restriction, the full 3×3 inertia matrix of the original algorithm reduces to a scalar z -axis inertia:

Algorithm 1 Planar inertia aggregation for the j th chain — planar reduction of Algorithm 1 in²

Require: CoG inertias $\tilde{I}_{z_{i_j}}$, masses m_{i_j} , relative positions f_{k_j} for $i = 2, \dots, N_j$

Ensure: Aggregated inertias ${}^0I_{z_{i_j}}$ for $i = 2, \dots, N_j$

```

1: Initialize:  ${}^0I_{z_{N_j}} \leftarrow \tilde{I}_{z_{N_j}}$ 
2: if  $2 \leq i < N_j$  then
3:   for  $k = N_j, \dots, i + 1$  do
4:      $\tilde{m}_{(k-1)_j} \leftarrow \sum_{b=k}^{N_j} m_{b_j}$  ▷ accumulated downstream mass
5:      ${}^0I_{z_{(k-1)_j}} \leftarrow {}^0I_{z_{k_j}} + \tilde{I}_{z_{(k-1)_j}} + \tilde{m}_{(k-1)_j} (f_{x,k_j}^2 + f_{y,k_j}^2)$  ▷ Steiner shift from frame  $k$  to
       frame  $k - 1$ 
6:   end for
7: end if
```

Here $\tilde{m}_{(k-1)_j} = \sum_{b=k}^{N_j} m_{b_j}$ is the total mass of all downstream bodies (from k to the end-effector), $\tilde{I}_{z_{(k-1)_j}}$ is the local inertia of body $k - 1$ about its own CoG, and $\tilde{m}_{(k-1)_j} (f_{x,k_j}^2 + f_{y,k_j}^2)$ is the Steiner correction that shifts the accumulated inertia from body k ’s CoG to body $(k - 1)$ ’s CoG. By the time the algorithm terminates, ${}^0I_{z_{2_j}}$ contains the inertia of the entire chain j

referred to the CoG of its first body.

The aggregated inertia of the base frame (body 1) applies the same shift one more time, from body 2's CoG to the base frame:

$${}^0I_{z_1} = \tilde{I}_{z_1} + \sum_{j=1}^3 \left({}^0I_{z_{2j}} + \tilde{m}_{1j} (f_{x,2j}^2 + f_{y,2j}^2) \right), \quad (2.2.18)$$

where $\tilde{m}_{1j} = \sum_{b=2}^{N_j} m_{b_j}$ is the total mass of chain j . This is the quantity denoted I_{z_0} in (2.2.22).

Expanding $\ddot{\theta}_{z_{i_j}}^0$ using (2.2.14):

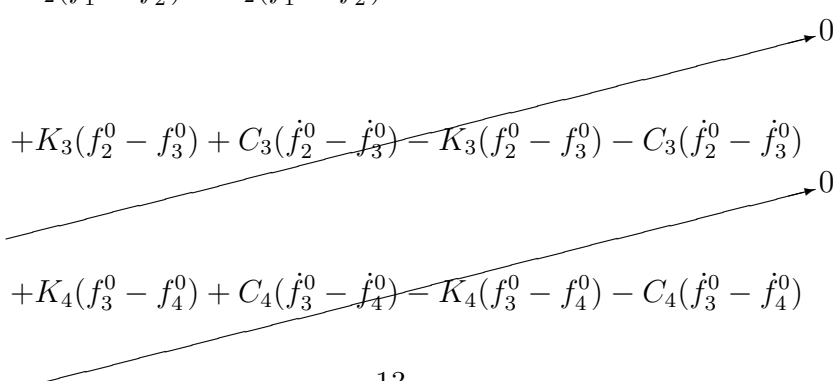
$$I_{z_{i_j}} \ddot{\theta}_{z_1}^0 + I_{z_{i_j}} \sum_{k=2}^i \ddot{\theta}_{z_{k_j}} = m_{z_{i_j}}^0 \quad (2.2.19)$$

2.2.1 For the base frame

To write the equations that govern the base frame, we start from Newton's equation $F = ma$. For every body ($i \geq 2$) within chain j , it connects to body $i - 1$ through a spring above and body $i + 1$ through the spring below:

$$\begin{aligned} m_{i_j} \ddot{f}_1^0 + m_{i_j} \sum_{k=2}^i \ddot{f}_{k_j} + m_{i_j} \sum_{k=2}^i \left[2 [\omega_{(k-1)_j}] \dot{f}_{k_j} + [\dot{\omega}_{(k-1)_j}] f_{k_j} + [\omega_{(k-1)_j}]^2 f_{k_j} \right] \\ - K_i (f_{(i-1)_j}^0 - f_{i_j}^0) - C_i (\dot{f}_{(i-1)_j}^0 - \dot{f}_{i_j}^0) + K_{i+1} (f_{i_j}^0 - f_{(i+1)_j}^0) + C_{i+1} (\dot{f}_{i_j}^0 - \dot{f}_{(i+1)_j}^0) = u_{i_j}^0 \end{aligned} \quad (2.2.20)$$

Summing (2.2.20) over all bodies in a chain (taking the optics chain $j = 2$, bodies $i = 2, 3, 4$ as a representative example), the internal spring forces cancel by Newton's third law:

$$\begin{aligned} & -K_2(f_1^0 - f_2^0) - C_2(\dot{f}_1^0 - \dot{f}_2^0) \\ & + K_3(f_2^0 - f_3^0) + C_3(\dot{f}_2^0 - \dot{f}_3^0) - K_3(f_2^0 - f_3^0) - C_3(\dot{f}_2^0 - \dot{f}_3^0) \\ & + K_4(f_3^0 - f_4^0) + C_4(\dot{f}_3^0 - \dot{f}_4^0) - K_4(f_3^0 - f_4^0) - C_4(\dot{f}_3^0 - \dot{f}_4^0) \end{aligned}$$


Only the boundary term $-K_2(f_1^0 - f_2^0) - C_2(\dot{f}_1^0 - \dot{f}_2^0)$ survives, which corresponds to the connection between body $i = 2$ and the base frame. This extends to any N_j -body chain. Recalling (2.2.3), $f_1^0 - f_{2_j}^0 = -f_{2_j}$, and following² with ${}^0K_{i_j} = {}^0R_{i_j}K_{i_j}{}^0R_{i_j}^T$ and ${}^0C_{i_j} = {}^0R_{i_j}C_{i_j}{}^0R_{i_j}^T$, the total translational equation of the base frame (absorbing the balance masses and collecting all chain contributions) is:

$$m_0 \ddot{f}_1^0 + {}^0\bar{C}_1 \dot{f}_1 + {}^0\bar{K}_1 f_1 - \sum_{j=1}^3 \left\{ {}^0\bar{C}_{2_j} \dot{f}_{2_j} + {}^0\bar{K}_{2_j} f_{2_j} \right\} = u_1^0 \quad (2.2.21)$$

where $m_0 = m_1 + \sum_{j=1}^3 \left(m_B^j + \sum_{i=2}^{N_j} m_{i_j} \right)$ is the total system mass, and the stiffness and damping matrices denote the 2×2 translational block.

By the exact same cancellation argument applied to the moment balance about the z -axis — the moment exerted by the spring connecting body i to body $i - 1$ is $-K_{i_j} \theta_{z_{i_j}}$, so internal moments cancel identically — the rotational equation of the base frame is:

$$I_{z_0} \ddot{\theta}_{z_1}^0 + {}^0\bar{C}_1 \dot{\theta}_{z_1} + {}^0\bar{K}_1 \theta_{z_1} - \sum_{j=1}^3 \left\{ {}^0\bar{C}_{2_j} \dot{\theta}_{z_{2_j}} + {}^0\bar{K}_{2_j} \theta_{z_{2_j}} \right\} = \tau_1^0 \quad (2.2.22)$$

where $I_{z_0} = I_{z_1} + \sum_{j=1}^3 \left(I_{z_B}^j + \sum_{i=2}^{N_j} I_{z_{i_j}} \right)$ is the total moment of inertia of the system about the z -axis, ${}^0\bar{C}_{i_j}$ and ${}^0\bar{K}_{i_j}$ are the scalar rotational damping and stiffness of the joint connecting body i to its predecessor in chain j , and τ_1^0 is the net applied torque on the base frame.

2.2.2 For each chain

Recalling (2.2.20) and (2.2.19), for the first body in the chain ($i = 2$), which connects to the base frame above and to body $i = 3$ below, the translational equation is:

$$\bar{m}_{2_j} \ddot{f}_{2_j}^0 + {}^0\bar{C}_{2_j} \dot{f}_{2_j} + {}^0\bar{K}_{2_j} f_{2_j} - \left\{ {}^0\bar{C}_{3_j} \dot{f}_{3_j} + {}^0\bar{K}_{3_j} f_{3_j} \right\} = u_{2_j}^0 \quad (2.2.23)$$

and the rotational equation is:

$$I_{z_{2_j}} \ddot{\theta}_{z_{2_j}}^0 + {}^0\bar{C}_{2_j} \dot{\theta}_{z_{2_j}} + {}^0\bar{K}_{2_j} \theta_{z_{2_j}} - {}^0\bar{C}_{3_j} \dot{\theta}_{z_{3_j}} - {}^0\bar{K}_{3_j} \theta_{z_{3_j}} = \tau_{2_j}^0 \quad (2.2.24)$$

For bodies $3 \leq i < N_j$, the same structure holds with bodies $2, \dots, i-1$ in between. The translational equation is:

$$\bar{m}_{i_j} \ddot{f}_{i_j}^0 + {}^0\bar{C}_{i_j} \dot{f}_{i_j} + {}^0\bar{K}_{i_j} f_{i_j} - \left\{ {}^0\bar{C}_{(i+1)_j} \dot{f}_{(i+1)_j} + {}^0\bar{K}_{(i+1)_j} f_{(i+1)_j} \right\} = u_{i_j}^0 \quad (2.2.25)$$

and the rotational equation is:

$$I_{z_{i_j}} \ddot{\theta}_{z_{i_j}}^0 + {}^0\bar{C}_{i_j} \dot{\theta}_{z_{i_j}} + {}^0\bar{K}_{i_j} \theta_{z_{i_j}} - {}^0\bar{C}_{(i+1)_j} \dot{\theta}_{z_{(i+1)_j}} - {}^0\bar{K}_{(i+1)_j} \theta_{z_{(i+1)_j}} = \tau_{i_j}^0 \quad (2.2.26)$$

Finally, for body $i = N_j$ (the end effector), there is no subsequent body in the chain. The translational and rotational equations are:

$$\bar{m}_{N_{j_j}} \ddot{f}_{N_{j_j}}^0 + {}^0\bar{C}_{N_{j_j}} \dot{f}_{N_{j_j}} + {}^0\bar{K}_{N_{j_j}} f_{N_{j_j}} = 0 \quad (2.2.27)$$

$$I_{z_{N_{j_j}}} \ddot{\theta}_{z_{N_{j_j}}}^0 + {}^0\bar{C}_{N_{j_j}} \dot{\theta}_{z_{N_{j_j}}} + {}^0\bar{K}_{N_{j_j}} \theta_{z_{N_{j_j}}} = 0 \quad (2.2.28)$$

2.2.3 State-space formulation

Because each body possesses three planar degrees of freedom (f_x, f_y, θ_z) along with their corresponding time derivatives, it contributes exactly six states to the system. Following the framework outlined in Appendix B of², the complete machine model comprises 6 bodies in the reticle chain ($N_1 = 7$), 3 bodies in the optics chain ($N_2 = 4$), and 6 bodies in the wafer chain ($N_3 = 7$). Including the common base frame (body 1), the system features a total of $1 + (N_1 - 1) + (N_2 - 1) + (N_3 - 1) = 16$ bodies. Consequently, the global state vector x encompasses $6 \times 16 = 96$ states:

$$x = \left[\underbrace{x_1, \dots, x_6}_{\text{base frame}}, \underbrace{x_7, \dots, x_{6N_1}}_{\text{chain 1: bodies } 2 \rightarrow N_1}, \underbrace{\quad}_{\text{chain 2}}, \underbrace{\quad}_{\text{chain 3}} \right]^T \quad (2.2.29)$$

The explicit mapping for each state is detailed in the table below. Here, p denotes the global sequential index of the i -th body in the j -th chain within the concatenated list of all bodies

(starting with the base frame, followed sequentially by chains 1, 2, and 3). It is crucial to note that every positional entry within the state vector represents a *relative* joint displacement; the corresponding absolute inertial position $f_{i_j}^0$ can be reconstructed using the recursive summations from (2.2.9) to (2.2.12).

Solving (2.2.21) and (2.2.22) for the base-frame accelerations yields:

$$\ddot{f}_1^0 = -\frac{{}^0\bar{C}_1}{m_0} \dot{f}_1 - \frac{{}^0\bar{K}_1}{m_0} f_1 + \frac{1}{m_0} \sum_{j=1}^3 \left\{ {}^0\bar{C}_{2_j} \dot{f}_{2_j} + {}^0\bar{K}_{2_j} f_{2_j} \right\} + \frac{u_1^0}{m_0} \quad (2.2.30)$$

$$\ddot{\theta}_{z_1}^0 = -\frac{{}^0\bar{C}_1}{I_{z_0}} \dot{\theta}_{z_1} - \frac{{}^0\bar{K}_1}{I_{z_0}} \theta_{z_1} + \frac{1}{I_{z_0}} \sum_{j=1}^3 \left\{ {}^0\bar{C}_{2_j} \dot{\theta}_{z_{2_j}} + {}^0\bar{K}_{2_j} \theta_{z_{2_j}} \right\} + \frac{\tau_1^0}{I_{z_0}} \quad (2.2.31)$$

Expressed in the standard matrix form $\dot{x} = Ax + Bu$, the first six rows governing the base-frame dynamics are:

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \\ \dot{x}_4 \\ \dot{x}_5 \\ \dot{x}_6 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & \dots \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 & \dots \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & \dots \\ -\frac{K_{1,xx}}{m_0} & -\frac{K_{1,xy}}{m_0} & 0 & -\frac{C_{1,xx}}{m_0} & -\frac{C_{1,xy}}{m_0} & 0 & \frac{K_{2_1,xx}}{m_0} & \frac{K_{2_1,xy}}{m_0} & 0 & \frac{C_{2_1,xx}}{m_0} & \dots \\ -\frac{K_{1,yx}}{m_0} & -\frac{K_{1,yy}}{m_0} & 0 & -\frac{C_{1,yx}}{m_0} & -\frac{C_{1,yy}}{m_0} & 0 & \frac{K_{2_1,yx}}{m_0} & \frac{K_{2_1,yy}}{m_0} & 0 & \frac{C_{2_1,yx}}{m_0} & \dots \\ 0 & 0 & -\frac{K_{1,zz}}{I_{z_0}} & 0 & 0 & -\frac{C_{1,zz}}{I_{z_0}} & 0 & 0 & \frac{K_{2_1,zz}}{I_{z_0}} & 0 & \dots \end{bmatrix} x + \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{m_0} \\ 0 \\ 0 \end{bmatrix} \quad (2.2.32)$$

General State-Space Form for the Chain Bodies

To systematically construct the global state-space matrices, we must reframe the equations of motion for the chain bodies in terms of the state vector variables. We map the physical planar coordinates of the p -th sequential body to the local state vectors: translational position $\mathbf{x}_{2p-1} = [x_{6p-5}, x_{6p-4}]^T$, translational velocity $\dot{\mathbf{x}}_{2p-1} = [x_{6p-2}, x_{6p-1}]^T$, rotational position $\mathbf{x}_{2p} = x_{6p-3}$, and rotational velocity $\dot{\mathbf{x}}_{2p} = x_{6p}$.

The state derivatives are dictated by the relative accelerations between adjacent bodies. By applying the kinematic identity $\ddot{f}_{i_j} = \ddot{f}_{i_j}^0 - \ddot{f}_{(i-1)_j}^0$ from (2.2.11), we determine the interactive

Table 2.2.1: State-space mapping for the global concatenated system vector

State-space Variable	Expression	Meaning
x_1	f_{x_1}	base frame x -position
x_2	f_{y_1}	base frame y -position
x_3	θ_{z_1}	base frame rotation
x_4	$\dot{f}_{x_1}$	base frame x -velocity
x_5	$\dot{f}_{y_1}$	base frame y -velocity
x_6	$\dot{\theta}_{z_1}$	base frame angular velocity
x_7	$f_{x_{2_1}}$	chain 1, body 2, x -position
x_8	$f_{y_{2_1}}$	chain 1, body 2, y -position
x_9	$\theta_{z_{2_1}}$	chain 1, body 2, rotation
x_{10}	$\dot{f}_{x_{2_1}}$	chain 1, body 2, x -velocity
x_{11}	$\dot{f}_{y_{2_1}}$	chain 1, body 2, y -velocity
x_{12}	$\dot{\theta}_{z_{2_1}}$	chain 1, body 2, angular velocity
$\vdots$	$\vdots$	$\vdots$
x_{6p-5}	$f_{x_{i_j}}$	chain j , body i , x -position
x_{6p-4}	$f_{y_{i_j}}$	chain j , body i , y -position
x_{6p-3}	$\theta_{z_{i_j}}$	chain j , body i , rotation
x_{6p-2}	$\dot{f}_{x_{i_j}}$	chain j , body i , x -velocity
x_{6p-1}	$\dot{f}_{y_{i_j}}$	chain j , body i , y -velocity
x_{6p}	$\dot{\theta}_{z_{i_j}}$	chain j , body i , angular velocity
$\vdots$	$\vdots$	$\vdots$

forces acting on each body within its parent's reference frame. Let p represent the global index of body i in chain j , while $p - 1$ and $p + 1$ denote the indices of its immediate upstream and downstream neighbors respectively (with the base frame positioned at $p = 1$).

For the **first body in each chain** ($i = 2$, index p , upstream index 1), the translational state derivatives ($\ddot{\mathbf{x}}_{2p-1} = \ddot{f}_{2j}$) and rotational state derivatives ($\ddot{\mathbf{x}}_{2p} = \ddot{\theta}_{z_{2j}}$) take the form:

$$\begin{aligned} \ddot{\mathbf{x}}_{2p-1} = & \frac{1}{\bar{m}_{2j}} \left[u_{2j}^0 - {}^0\bar{C}_{2j} \dot{\mathbf{x}}_{2p-1} - {}^0\bar{K}_{2j} \mathbf{x}_{2p-1} + {}^0\hat{C}_{3j} \dot{\mathbf{x}}_{2p+1} + {}^0\hat{K}_{3j} \mathbf{x}_{2p+1} \right] \\ & - \frac{1}{m_0} \left[u_1^0 - {}^0\bar{C}_1 \dot{\mathbf{x}}_1 - {}^0\bar{K}_1 \mathbf{x}_1 + \sum_{k=1}^3 \left({}^0\hat{C}_{2k} \dot{\mathbf{x}}_{2p_{2k}-1} + {}^0\hat{K}_{2k} \mathbf{x}_{2p_{2k}-1} \right) \right] \end{aligned} \quad (2.2.33)$$

$$\begin{aligned} \ddot{\mathbf{x}}_{2p} = & \frac{1}{I_{z_{2j}}} \left[\tau_{2j}^0 - {}^0\bar{c}_{\theta,2j} \dot{\mathbf{x}}_{2p} - {}^0\bar{k}_{\theta,2j} \mathbf{x}_{2p} + {}^0\hat{c}_{\theta,3j} \dot{\mathbf{x}}_{2p+2} + {}^0\hat{k}_{\theta,3j} \mathbf{x}_{2p+2} \right] \\ & - \frac{1}{I_{z_0}} \left[\tau_1^0 - {}^0\bar{C}_1 \dot{\mathbf{x}}_2 - {}^0\bar{K}_1 \mathbf{x}_2 + \sum_{k=1}^3 \left({}^0\hat{c}_{\theta,2k} \dot{\mathbf{x}}_{2p_{2k}} + {}^0\hat{k}_{\theta,2k} \mathbf{x}_{2p_{2k}} \right) \right] \end{aligned} \quad (2.2.34)$$

For a **general intermediate body** ($2 < i < N_j$, index p), the dynamics are shaped by upstream, local, and downstream interactions:

$$\begin{aligned} \ddot{\mathbf{x}}_{2p-1} = & \frac{1}{\bar{m}_{i_j}} \left[u_{i_j}^0 - {}^0\bar{C}_{i_j} \dot{\mathbf{x}}_{2p-1} - {}^0\bar{K}_{i_j} \mathbf{x}_{2p-1} + {}^0\hat{C}_{(i+1)_j} \dot{\mathbf{x}}_{2p+1} + {}^0\hat{K}_{(i+1)_j} \mathbf{x}_{2p+1} \right] \\ & - \frac{1}{\bar{m}_{(i-1)_j}} \left[u_{(i-1)_j}^0 - {}^0\bar{C}_{(i-1)_j} \dot{\mathbf{x}}_{2p-3} - {}^0\bar{K}_{(i-1)_j} \mathbf{x}_{2p-3} + {}^0\hat{C}_{i_j} \dot{\mathbf{x}}_{2p-1} + {}^0\hat{K}_{i_j} \mathbf{x}_{2p-1} \right] \end{aligned} \quad (2.2.35)$$

$$\begin{aligned} \ddot{\mathbf{x}}_{2p} = & \frac{1}{I_{z_{i_j}}} \left[\tau_{i_j}^0 - {}^0\bar{c}_{\theta,i_j} \dot{\mathbf{x}}_{2p} - {}^0\bar{k}_{\theta,i_j} \mathbf{x}_{2p} + {}^0\hat{c}_{\theta,(i+1)_j} \dot{\mathbf{x}}_{2p+2} + {}^0\hat{k}_{\theta,(i+1)_j} \mathbf{x}_{2p+2} \right] \\ & - \frac{1}{I_{z_{(i-1)_j}}} \left[\tau_{(i-1)_j}^0 - {}^0\bar{c}_{\theta,(i-1)_j} \dot{\mathbf{x}}_{2p-2} - {}^0\bar{k}_{\theta,(i-1)_j} \mathbf{x}_{2p-2} + {}^0\hat{c}_{\theta,i_j} \dot{\mathbf{x}}_{2p} + {}^0\hat{k}_{\theta,i_j} \mathbf{x}_{2p} \right] \end{aligned} \quad (2.2.36)$$

Finally, for the **end-effector body** ($i = N_j$, index p), the downstream force terms naturally

vanish as there are no subsequent bodies:

$$\begin{aligned}\ddot{\mathbf{x}}_{2p-1} &= \frac{1}{\bar{m}_{N_{jj}}} \left[u_{N_{jj}}^0 - {}^0\bar{C}_{N_{jj}} \dot{\mathbf{x}}_{2p-1} - {}^0\bar{K}_{N_{jj}} \mathbf{x}_{2p-1} \right] \\ &\quad - \frac{1}{\bar{m}_{(N_j-1)_j}} \left[u_{(N_j-1)_j}^0 - {}^0\bar{C}_{(N_j-1)_j} \dot{\mathbf{x}}_{2p-3} - {}^0\bar{K}_{(N_j-1)_j} \mathbf{x}_{2p-3} + {}^0\hat{C}_{N_{jj}} \dot{\mathbf{x}}_{2p-1} + {}^0\hat{K}_{N_{jj}} \mathbf{x}_{2p-1} \right]\end{aligned}\tag{2.2.37}$$

$$\begin{aligned}\ddot{\mathbf{x}}_{2p} &= \frac{1}{I_{z_{N_{jj}}}} \left[\tau_{N_{jj}}^0 - {}^0\bar{c}_{\theta, N_{jj}} \dot{\mathbf{x}}_{2p} - {}^0\bar{k}_{\theta, N_{jj}} \mathbf{x}_{2p} \right] \\ &\quad - \frac{1}{I_{z_{(N_j-1)_j}}} \left[\tau_{(N_j-1)_j}^0 - {}^0\bar{c}_{\theta, (N_j-1)_j} \dot{\mathbf{x}}_{2p-2} - {}^0\bar{k}_{\theta, (N_j-1)_j} \mathbf{x}_{2p-2} + {}^0\hat{c}_{\theta, N_{jj}} \dot{\mathbf{x}}_{2p} + {}^0\hat{k}_{\theta, N_{jj}} \mathbf{x}_{2p} \right]\end{aligned}\tag{2.2.38}$$

Chapter 3

Control Objectives

3.1 Introduction

As we established before, the accurate relative positioning between the reticle and the wafer is crucial for *step-and-scan* machines to produce functional ICs. To analyze these positioning requirements in a multi-body framework, Al-Rawashdeh et al² propose modeling the lithography scanner as a combination of three open-loop serial chains: the reticle chain ($j = 1$, blue), the optics chain ($j = 2$, purple), and the wafer chain ($j = 3$, red), the latter being of primary interest for the *step-and-scan* motion problem studied here.

Chapter 4

Methodology

4.1 Introduction

Trajectory generation is a well studied problem. We understand that trajectories that have smooth higher order derivatives result in less excitation of poorly damped modes in the structure. To properly test the trajectories we will be generating against the yield they will generate on the machine, then we need a test bed on which they can be performed on-line before actually reproducing them in the machine.

4.2 Simulation

We implemented a simulation of the model we recovered from the state-space characterization article² using the MuJoCo⁸ physics engine for Python. The simulation serves as a virtual test bed where the trajectories proposed can be validated under the dynamic model we developed in chapter 2.

The MuJoCo Python bindings allows us to define the context of our simulation from an XML string by calling the `mujoco.MjModel.from_xml_string(MODEL_XML)` method. We extend from

MuJoCo’s `<worldbody>` tag on which we place a root `<body>` tag, representing our *base frame*; this allows us to organize the machine in the simulation using the three chain hierarchical tree. Table 4.2.1 maps the specific names used in the MuJoCo XML to their counterparts in the structural diagram of Figure 2.1.

Table 4.2.1: Mapping of MuJoCo XML bodies to the model components

<code><body></code> name	property	Model body
<code>base_frame</code>		Base Frame ($\mathcal{F}_1$)
<code>w_ls_act</code>		Wafer LS ($i_3 = 2$)
<code>w_ls_flex</code>		flex coupling ($i_3 = 3$)
<code>w_ss_act</code>		Wafer SS ($i_3 = 4$)
<code>w_ss_flex</code>		flex coupling ($i_3 = 5$)
<code>w_st_act</code>		Wafer Stage ($i_3 = 6$)
<code>wafer</code>		Die ($N_3 = 7$)
<code>r_ls_act</code>		Reticle LS ($i_1 = 2$)
<code>r_ls_flex</code>		flex coupling ($i_1 = 3$)
<code>r_ss_act</code>		Reticle SS ($i_1 = 4$)
<code>r_ss_flex</code>		flex coupling ($i_1 = 5$)
<code>r_st_act</code>		Reticle Stage ($i_1 = 6$)
<code>mask</code>		Mask ($N_1 = 7$)
<code>metro_frame</code>		Metrology Frame ($i_2 = 2$)
<code>optics_box</code>		Optics Box ($i_2 = 3$)
<code>lens</code>		Lens Element ($i_2 = 4$)

The simulation environment is defined through several specialized XML attributes that handle different aspects of the physical world. Table 4.2.2 summarizes the function of each tag within the context of our lithography scanner model.

Table 4.2.2: MuJoCo XML tag functions in the scanner simulation

Tag	Function
<code><worldbody></code>	Defines the inertial root, the floor plane ($\mathcal{F}_0$).
<code><body></code>	Creates a rigid body with its own local coordinate system.
<code><inertial></code>	Assigns mass m_{i_j} and the aggregated inertia tensor I_{i_j} .
<code><joint></code>	Implements the degrees of freedom f_{i_j} and $\theta_{z_{i_j}}$.
<code><geom></code>	Defines the material (granite, silicon, glass) and collision shapes.
<code><actuator></code>	Models the Lorentz and moving-coil motors via servo controllers.

The mechanical properties of the couplings, represented by the Kelvin-Voigt elements, are implemented via the `stiffness` and `damping` attributes of the `<joint>` tags. For actuated stages, where the physical stiffness is replaced by the controller’s active response, the simulation uses high `kp` values. Tables 4.2.3, 4.2.4, and 4.2.5 illustrate the distribution of these variables for the three chains.

Table 4.2.3: Wafer chain stiffness (K) and damping (C) parameters

Coupling Interface	Stiffness Variable (K)	Damping Variable (C)
Floor $\rightarrow$ Base Frame	K1	C1
Base $\rightarrow$ LS Stage	KP_ACT	KV_ACT_TRANS
LS Stage $\rightarrow$ LS Flex	K_LS_F	C_LS_F
SS Stage $\rightarrow$ SS Flex	K_SS_F	C_SS_F
Fine Stage $\rightarrow$ Flex	K_ST_F	C_ST_F
Stage $\rightarrow$ Die	K8_W	C8_W

As we mentioned, the simulation topology mirrors the multi-body hierarchy described in Chapter 2. The common base frame ($\mathcal{F}_1$) is anchored to the world frame ($\mathcal{F}_0$) via a 3-DOF joint (two prismatic, one revolute) representing the isolation system with stiffness K_1 and damping

Table 4.2.4: Reticle chain stiffness (K) and damping (C) parameters

Coupling Interface	Stiffness Variable (K)	Damping Variable (C)
Floor $\rightarrow$ Base Frame	K1	C1
Base $\rightarrow$ LS Stage	KP_ACT	KV_ACT_TRANS
LS Stage $\rightarrow$ LS Flex	K_LS_F	C_LS_F
SS Stage $\rightarrow$ SS Flex	K_SS_F	C_SS_F
Fine Stage $\rightarrow$ Flex	K_ST_F	C_ST_F
Stage $\rightarrow$ Mask	K8_R	C8_R

Table 4.2.5: Optics chain stiffness (K) and damping (C) parameters

Coupling Interface	Stiffness Variable (K)	Damping Variable (C)
Base Frame $\rightarrow$ Metro Frame	K2_0	C2_0
Metro Frame $\rightarrow$ Optics Box	K3_0	C3_0
Optics Box $\rightarrow$ Lens Element	K4_0	C4_0

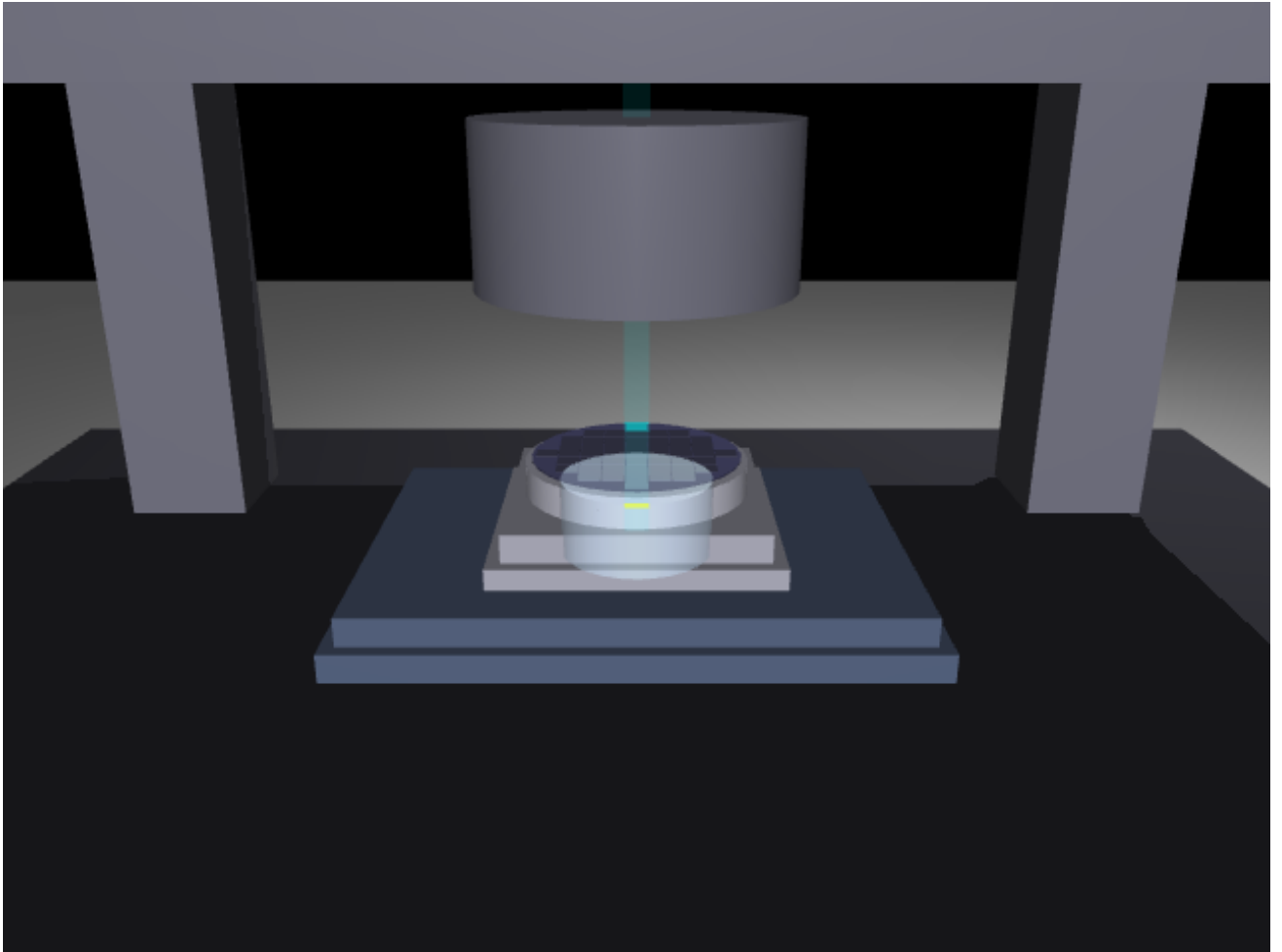


Figure 4.1: Screenshot of the MuJoCo simulation in the midst of the scanning motion, showing the wafer and the optics chain under the metrology frame

C_1 as seen in Eq. (2.2.21) and (2.2.22). From this base, three serial chains are constructed using nested body definitions in MuJoCo’s XML format, where each child body’s joint coordinate corresponds directly to the relative displacement f_{i_j} and rotation $\theta_{z_{i_j}}$ defined in our kinematics model.

The passive Kelvin-Voigt couplings ($K_{i,j}, C_{i,j}$) are implemented as MuJoCo joints with explicit stiffness and damping parameters. For the actuated stages (Long-Stroke, Short-Stroke, and Fine stages), we use MuJoCo’s position actuators. To approximate the near-infinite stiffness of a rigid servo motor while maintaining numerical stability with the implicit integrator at a timestep of $dt = 10^{-4}$ s, we employ a proportional gain $K_P = 10^7$, with derivative gains K_V tuned to the C/K ratios provided in the characterization data.

The simulation incorporates a *StepperController* that generates the references for the scanning and stepping phases. During the scanning phase, the wafer stage moves at a constant velocity $v_{w,scan} = 0.8$ m/s while the reticle stage moves in the opposite direction at $v_{r,scan} = 3.2$ m/s, maintaining the demagnification ratio $\alpha = 0.25$. This testing framework allows us to evaluate the impact of the centripetal and Coriolis terms from Eq. (2.2.11) and the effectiveness of the inertia aggregation described in Algorithm 1 when subjected to high-acceleration ($a_{w,max} = 20$ m/s²) transients.

Conclusion

En este trabajo de investigacion se ha propuesto un esquema de ...

Bibliography

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Appendix A

Full-State Space Model

A.1 State-Space Equations for the Wafer Chain

The wafer chain corresponds to $j = 3$, with $N_3 = 7$ (bodies $i = 2, \dots, 7$). These map to global state vectors $\mathbf{x}_{2p-1}$ (translational) and $\mathbf{x}_{2p}$ (rotational) for $p = 12, \dots, 17$. The state-space equations are represented below.

First body of the wafer chain ($i = 2, p = 12$):

$$\begin{aligned} \ddot{\mathbf{x}}_{23} = & \frac{1}{\bar{m}_{23}} \left[u_{23}^0 - {}^0\bar{C}_{23} \dot{\mathbf{x}}_{23} - {}^0\bar{K}_{23} \mathbf{x}_{23} + {}^0\hat{C}_{33} \dot{\mathbf{x}}_{25} + {}^0\hat{K}_{33} \mathbf{x}_{25} \right] \\ & - \frac{1}{m_0} \left[u_1^0 - {}^0\bar{C}_1 \dot{\mathbf{x}}_1 - {}^0\bar{K}_1 \mathbf{x}_1 + \sum_{k=1}^3 \left({}^0\hat{C}_{2k} \dot{\mathbf{x}}_{2p_{2k}-1} + {}^0\hat{K}_{2k} \mathbf{x}_{2p_{2k}-1} \right) \right] \end{aligned} \quad (\text{A.1.1})$$

$$\begin{aligned} \ddot{\mathbf{x}}_{24} = & \frac{1}{I_{z_{23}}} \left[\tau_{23}^0 - {}^0\bar{c}_{\theta,23} \dot{\mathbf{x}}_{24} - {}^0\bar{k}_{\theta,23} \mathbf{x}_{24} + {}^0\hat{c}_{\theta,33} \dot{\mathbf{x}}_{26} + {}^0\hat{k}_{\theta,33} \mathbf{x}_{26} \right] \\ & - \frac{1}{I_{z_0}} \left[\tau_1^0 - {}^0\bar{C}_1 \dot{\mathbf{x}}_2 - {}^0\bar{K}_1 \mathbf{x}_2 + \sum_{k=1}^3 \left({}^0\hat{c}_{\theta,2k} \dot{\mathbf{x}}_{2p_{2k}} + {}^0\hat{k}_{\theta,2k} \mathbf{x}_{2p_{2k}} \right) \right] \end{aligned} \quad (\text{A.1.2})$$

Intermediate body ($i = 3, p = 13$):

$$\begin{aligned} \ddot{\mathbf{x}}_{25} = & \frac{1}{\bar{m}_{33}} \left[u_{33}^0 - {}^0\bar{C}_{33} \dot{\mathbf{x}}_{25} - {}^0\bar{K}_{33} \mathbf{x}_{25} + {}^0\hat{C}_{(4)3} \dot{\mathbf{x}}_{27} + {}^0\hat{K}_{(4)3} \mathbf{x}_{27} \right] \\ & - \frac{1}{\bar{m}_{(2)3}} \left[u_{(2)3}^0 - {}^0\bar{C}_{(2)3} \dot{\mathbf{x}}_{23} - {}^0\bar{K}_{(2)3} \mathbf{x}_{23} + {}^0\hat{C}_{33} \dot{\mathbf{x}}_{25} + {}^0\hat{K}_{33} \mathbf{x}_{25} \right] \end{aligned} \quad (\text{A.1.3})$$

$$\begin{aligned} \ddot{\mathbf{x}}_{26} = & \frac{1}{I_{z_{33}}} \left[\tau_{33}^0 - {}^0\bar{c}_{\theta,33} \dot{\mathbf{x}}_{26} - {}^0\bar{k}_{\theta,33} \mathbf{x}_{26} + {}^0\hat{c}_{\theta,(4)3} \dot{\mathbf{x}}_{28} + {}^0\hat{k}_{\theta,(4)3} \mathbf{x}_{28} \right] \\ & - \frac{1}{I_{z_{(2)3}}} \left[\tau_{(2)3}^0 - {}^0\bar{c}_{\theta,(2)3} \dot{\mathbf{x}}_{24} - {}^0\bar{k}_{\theta,(2)3} \mathbf{x}_{24} + {}^0\hat{c}_{\theta,33} \dot{\mathbf{x}}_{26} + {}^0\hat{k}_{\theta,33} \mathbf{x}_{26} \right] \end{aligned} \quad (\text{A.1.4})$$

Intermediate body ($i = 4, p = 14$):

$$\begin{aligned}\ddot{\mathbf{x}}_{27} = & \frac{1}{\bar{m}_{43}} \left[u_{43}^0 - {}^0\bar{C}_{43} \dot{\mathbf{x}}_{27} - {}^0\bar{K}_{43} \mathbf{x}_{27} + {}^0\hat{C}_{(5)3} \dot{\mathbf{x}}_{29} + {}^0\hat{K}_{(5)3} \mathbf{x}_{29} \right] \\ & - \frac{1}{\bar{m}_{(3)3}} \left[u_{(3)3}^0 - {}^0\bar{C}_{(3)3} \dot{\mathbf{x}}_{25} - {}^0\bar{K}_{(3)3} \mathbf{x}_{25} + {}^0\hat{C}_{43} \dot{\mathbf{x}}_{27} + {}^0\hat{K}_{43} \mathbf{x}_{27} \right]\end{aligned}\quad (\text{A.1.5})$$

$$\begin{aligned}\ddot{\mathbf{x}}_{28} = & \frac{1}{I_{z_{43}}} \left[\tau_{43}^0 - {}^0\bar{c}_{\theta,43} \dot{\mathbf{x}}_{28} - {}^0\bar{k}_{\theta,43} \mathbf{x}_{28} + {}^0\hat{c}_{\theta,(5)3} \dot{\mathbf{x}}_{30} + {}^0\hat{k}_{\theta,(5)3} \mathbf{x}_{30} \right] \\ & - \frac{1}{I_{z_{(3)3}}} \left[\tau_{(3)3}^0 - {}^0\bar{c}_{\theta,(3)3} \dot{\mathbf{x}}_{26} - {}^0\bar{k}_{\theta,(3)3} \mathbf{x}_{26} + {}^0\hat{c}_{\theta,43} \dot{\mathbf{x}}_{28} + {}^0\hat{k}_{\theta,43} \mathbf{x}_{28} \right]\end{aligned}\quad (\text{A.1.6})$$

Intermediate body ($i = 5, p = 15$):

$$\begin{aligned}\ddot{\mathbf{x}}_{29} = & \frac{1}{\bar{m}_{53}} \left[u_{53}^0 - {}^0\bar{C}_{53} \dot{\mathbf{x}}_{29} - {}^0\bar{K}_{53} \mathbf{x}_{29} + {}^0\hat{C}_{(6)3} \dot{\mathbf{x}}_{31} + {}^0\hat{K}_{(6)3} \mathbf{x}_{31} \right] \\ & - \frac{1}{\bar{m}_{(4)3}} \left[u_{(4)3}^0 - {}^0\bar{C}_{(4)3} \dot{\mathbf{x}}_{27} - {}^0\bar{K}_{(4)3} \mathbf{x}_{27} + {}^0\hat{C}_{53} \dot{\mathbf{x}}_{29} + {}^0\hat{K}_{53} \mathbf{x}_{29} \right]\end{aligned}\quad (\text{A.1.7})$$

$$\begin{aligned}\ddot{\mathbf{x}}_{30} = & \frac{1}{I_{z_{53}}} \left[\tau_{53}^0 - {}^0\bar{c}_{\theta,53} \dot{\mathbf{x}}_{30} - {}^0\bar{k}_{\theta,53} \mathbf{x}_{30} + {}^0\hat{c}_{\theta,(6)3} \dot{\mathbf{x}}_{32} + {}^0\hat{k}_{\theta,(6)3} \mathbf{x}_{32} \right] \\ & - \frac{1}{I_{z_{(4)3}}} \left[\tau_{(4)3}^0 - {}^0\bar{c}_{\theta,(4)3} \dot{\mathbf{x}}_{28} - {}^0\bar{k}_{\theta,(4)3} \mathbf{x}_{28} + {}^0\hat{c}_{\theta,53} \dot{\mathbf{x}}_{30} + {}^0\hat{k}_{\theta,53} \mathbf{x}_{30} \right]\end{aligned}\quad (\text{A.1.8})$$

Intermediate body ($i = 6, p = 16$):

$$\begin{aligned}\ddot{\mathbf{x}}_{31} = & \frac{1}{\bar{m}_{63}} \left[u_{63}^0 - {}^0\bar{C}_{63} \dot{\mathbf{x}}_{31} - {}^0\bar{K}_{63} \mathbf{x}_{31} + {}^0\hat{C}_{(7)3} \dot{\mathbf{x}}_{33} + {}^0\hat{K}_{(7)3} \mathbf{x}_{33} \right] \\ & - \frac{1}{\bar{m}_{(5)3}} \left[u_{(5)3}^0 - {}^0\bar{C}_{(5)3} \dot{\mathbf{x}}_{29} - {}^0\bar{K}_{(5)3} \mathbf{x}_{29} + {}^0\hat{C}_{63} \dot{\mathbf{x}}_{31} + {}^0\hat{K}_{63} \mathbf{x}_{31} \right]\end{aligned}\quad (\text{A.1.9})$$

$$\begin{aligned}\ddot{\mathbf{x}}_{32} = & \frac{1}{I_{z_{63}}} \left[\tau_{63}^0 - {}^0\bar{c}_{\theta,63} \dot{\mathbf{x}}_{32} - {}^0\bar{k}_{\theta,63} \mathbf{x}_{32} + {}^0\hat{c}_{\theta,(7)3} \dot{\mathbf{x}}_{34} + {}^0\hat{k}_{\theta,(7)3} \mathbf{x}_{34} \right] \\ & - \frac{1}{I_{z_{(5)3}}} \left[\tau_{(5)3}^0 - {}^0\bar{c}_{\theta,(5)3} \dot{\mathbf{x}}_{30} - {}^0\bar{k}_{\theta,(5)3} \mathbf{x}_{30} + {}^0\hat{c}_{\theta,63} \dot{\mathbf{x}}_{32} + {}^0\hat{k}_{\theta,63} \mathbf{x}_{32} \right]\end{aligned}\quad (\text{A.1.10})$$

End-effector body ($i = 7, p = 17$):

$$\begin{aligned}\ddot{\mathbf{x}}_{33} = & \frac{1}{\bar{m}_{73}} \left[u_{73}^0 - {}^0\bar{C}_{73} \dot{\mathbf{x}}_{33} - {}^0\bar{K}_{73} \mathbf{x}_{33} \right] \\ & - \frac{1}{\bar{m}_{(6)3}} \left[u_{(6)3}^0 - {}^0\bar{C}_{(6)3} \dot{\mathbf{x}}_{31} - {}^0\bar{K}_{(6)3} \mathbf{x}_{31} + {}^0\hat{C}_{73} \dot{\mathbf{x}}_{33} + {}^0\hat{K}_{73} \mathbf{x}_{33} \right]\end{aligned}\quad (\text{A.1.11})$$

$$\begin{aligned}\ddot{\mathbf{x}}_{34} = & \frac{1}{I_{z_{73}}} \left[\tau_{73}^0 - {}^0\bar{c}_{\theta,73} \dot{\mathbf{x}}_{34} - {}^0\bar{k}_{\theta,73} \mathbf{x}_{34} \right] \\ & - \frac{1}{I_{z_{(6)3}}} \left[\tau_{(6)3}^0 - {}^0\bar{c}_{\theta,(6)3} \dot{\mathbf{x}}_{32} - {}^0\bar{k}_{\theta,(6)3} \mathbf{x}_{32} + {}^0\hat{c}_{\theta,73} \dot{\mathbf{x}}_{34} + {}^0\hat{k}_{\theta,73} \mathbf{x}_{34} \right]\end{aligned}\quad (\text{A.1.12})$$